Temporal Feature Engineering Pipeline: 10-Step Sliding Window

5 Raw Metrics

(Continuous KPIs)

1. CPU Utilization (%)
2. Memory Usage (%)
3. Disk Usage (%)
4. Net Bytes Sent (MB/s)
5. Net Bytes Recv (MB/s)

10-Step Sliding Window

Trailing FIFO Buffer: [t-9 .. t]

Strict Leak-Free Construction:
Zero future visibility ($t' \leq t$)



Short Window (w=5) | Full Window (w=10)

StandardScaler Normalization
(Fit exclusively on training split)

45 Derived Temporal Features

(9 features per metric × 5 metrics)

- **Current Value (x_t)**
  Point-in-time state
- **Rolling Means (μ_5, μ_{10})**
  Local & historical level
- **Rolling Deltas (Δ_5, Δ_{10})**
  Magnitude shift from baseline
- **Rolling Slopes (m_5, m_{10})**
  OLS linear trend trajectory
- **Rolling Std Dev (σ_5, σ_{10})**
  Incidence of temporal jitter

Forward-looking label target: y = 1 if failure occurs in [t+1 .. t+10]. Current step t excluded from target to eliminate circularity.